

Revision Notes for Class 7 Maths Chapter 4 Expressions using Letter-Numbers

Mathematical expressions use letters alongside numbers to describe relations, patterns, and unknown values in a simple form. This chapter introduces the idea of using letters like a , s , n , c , or j to replace objects or numbers in real-life situations, allowing concise representation and easy calculation of different scenarios.

Understanding Letter-Numbers and Expressions

When a relationship is described in words, it can usually be expressed in a mathematical form using letters. For example, if Shabnam is always 3 years older than Aftab, then her age will be Aftab's age plus 3. Using a for Aftab's age and s for Shabnam's age, the relation becomes $s = a + 3$. These letters act as variables, called letter-numbers, and together form algebraic expressions.

Sometimes, knowing one value, we can find the other. If $a = 23$, then $s = 23 + 3 = 26$ years. Similarly, if we know Shabnam's age, Aftab's age can be found by $a = s - 3$. These methods make calculations flexible and quick for any given value.

Patterns and Algebraic Expressions

Letters can represent patterns, such as building objects or dealing with money. For example, to make a pattern of L shapes using matchsticks, if one L requires 2 matchsticks, then n Ls will need $2 \times n$ matchsticks. So, the expression for the number of sticks becomes $2n$. This is especially useful when you want to know the number for any value of n .

Everyday Uses of Algebraic Expressions

Algebraic expressions are not limited to ages or shapes. They help in calculating costs, measuring lengths, and more. For example, if Ketaki buys c coconuts (₹35 each) and j kilograms of jaggery (₹60 per kg), the total cost is $(c \times 35) + (j \times 60)$. Expressions like these are useful in shopping, budgeting, or making business estimations.

You may also use expressions to find perimeters of shapes. For a square with each side q , the perimeter is $4q$. For regular polygons, multiply the number of sides by the given side length. Such formulas help in geometry problems or when building things.

Solving Real-life Situations

Algebraic expressions are very helpful in daily activities. If someone has a 20 m long pipe and joins another pipe of length k , the total length is $20 + k$. In dealing with money, consider a table showing the number of ₹100, ₹20, and ₹5 notes. If you have x , y , and z notes of each, a general expression for total value could be $100x + 20y + 5z$.

Time calculations can also be expressed. For example, time taken to grind y kilograms of flour may be $10 + 8y$ minutes, meaning as y increases, time increases linearly.

Writing Expressions for Phrases

Some common English phrases can directly convert into mathematical expressions:

- 5 more than a number: $n + 5$
- 4 less than a number: $n - 4$
- 2 less than thirteen times a number: $13n - 2$
- 13 less than two times a number: $2n - 13$

Similarly, expressions like $8x + 3y$ or $15j - 2k$ can describe real situations, such as total costs or the number of items.

Working with Arithmetic and Algebraic Expressions

Operations used in arithmetic are the same in algebra. You can add, subtract, multiply, or use brackets in expressions. For example, to calculate $23 - 10 \times 2$, first do the multiplication. In algebra, always follow the correct order of operations (BODMAS).

Simplification is key. For example, the perimeter of a rectangle with sides l and b is $l + b + l + b = 2l + 2b$. This makes calculations faster, especially when specific values are known.

Use the simpler form of multiplication where possible. Instead of writing $4 \times n$, you can simply write $4n$. This helps to keep expressions short and clear.

Common Practice Problems

Students should practice by finding values for given variables. For example, if $m = 2$, the value of $5m + 3$ is 13. Mistakes can happen, so always substitute correctly and follow the right signs.

You may also face problems where you need to combine and simplify terms, especially in money calculations or sales tables. For example:

Day	Pencils (Price c)	Erasers (Price d)
1	5	4
2	3	6
3	10	1

Add up pencils: $5c + 3c + 10c = 18c$. Erasers: $4d + 6d + 1d = 11d$. In total: $18c + 11d$.

Application with Patterns and Calendars

Expressions help to discover and predict patterns, like matchstick figures or repeating sequences. For example, if you keep adding an L-shape with 2 matchsticks, at each step y the total matchsticks are $2y + 1$.

Patterns also appear in calendars. For any 2×2 block in a calendar, the sum of diagonals remains equal: such as $a + (a+8) = 2a + 8$. Such relationships are efficiently written and explored using algebra.

Summary of Key Points

- Letters, called variables or letter-numbers, help describe general situations in maths using algebraic expressions.
- Algebraic expressions are useful for representing patterns, calculating for any value, and simplifying word problems.
- The rules for arithmetic operations (addition, multiplication etc.) apply equally to expressions with letters.
- Expressions can be easily evaluated by substituting different values for the letters.
- Writing expressions saves time and effort in repeated calculations or exploring patterns and relationships in maths and life.